1. main.c

```c
#include <stdio.h>
#include "header.h"
#include <string.h>
#include <stdlib.h>

#define MAX_NUMBERS 1000

int main(){
  int question;
  printf("Enter Question Number: ");
  scanf("%d", &question);
  /*
    Clear the input buffer to handle the newline character after scanf("%d", &question)
  */
  getchar();
  switch(question){
    case 1:{
      student students[MAX_NUMBERS];
      int numStudents = 0;

      printf("Enter the number of students: ");
      scanf("%d", &numStudents);
      getchar();
      for (int i = 0; i < numStudents; i++) {
        printf("\nEnter details for student %d:\n", i + 1);

        printf("Roll Number: ");
        scanf("%d", &students[i].rollno);
        getchar();
        printf("First Name: ");
```

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            scanf("%s", students[i].name.firstName);

            printf("Middle Name: ");
            scanf("%s", students[i].name.middleName);

            printf("Last Name: ");
            scanf("%s", students[i].name.lastName);

            printf("Gender: ");
            scanf("%s", students[i].gender);

            printf("Date of Birth (dd mm yyyy): ");
            scanf("%d %d %d", &students[i].dob.day, &students[i].dob.month, &students[i].dob.year);

            printf("Marks (Mathematics Science Computer Science): ");
            scanf("%d %d %d", &students[i].marks.maths, &students[i].marks.science,
&students[i].marks.computerScience);
          }
          printf("\n");
          for (int i = 0; i < numStudents; i++) {
            float aggregatePercentage = calculateAggregatePercentage(&students[i]);
            if (aggregatePercentage < 40) {
              printf("This student has less than 40%% aggregate and aggregate marks are %f:\n",
aggregatePercentage);
              displayStudent(&students[i]);
            }
            printf("\n");
          }
          break;
        }
        case 2:{
          hotel hotels[MAX_NUMBERS];
```

```
int numHotels = 0;
printf("Enter the number of hotels: ");
scanf("%d", &numHotels);
getchar();
for (int i = 0; i < numHotels; i++) {
   printf("\nEnter details for hotel %d:\n", i + 1);
   printf("Hotel Name: ");
   scanf("%[^\n]s", hotels[i].name);
   getchar();
   printf("Address: ");
   scanf("%[^\n]s", hotels[i].address);
   printf("Grade: ");
   scanf("%d", &hotels[i].grade);
   printf("Number Of Rooms: ");
   scanf("%d", &hotels[i].number_of_rooms);
   printf("Room Charges: ");
   scanf("%d", &hotels[i].room_charges);
   getchar();
}
int g;
printf("Enter the grade of which hotels you want:");
scanf("%d", &g);
for(int i = 0; i < numHotels; i++){
   hotelsInParticularGrade(&hotels[i], g);
}
int value;
printf("Enter the value below which rooms in hotels you want:");
scanf("%d", &value);
for(int i = 0; i < numHotels; i++){
   hotelsOfParticularValue(&hotels[i], value);
}
```

```
        break;
    }
    case 3:{
        time start_time, end_time;
        printf("Enter start time (hr min sec): ");
        scanf("%d %d %d", &start_time.hr, &start_time.min, &start_time.sec);
        printf("Enter end time (hr min sec): ");
        scanf("%d %d %d", &end_time.hr, &end_time.min, &end_time.sec);

        printf("\n");
        while (!(start_time.hr == end_time.hr && start_time.min == end_time.min && start_time.sec
== end_time.sec)) {
            printf("GOOD DAY\n");
            increment_time(&start_time);
        }
        break;
    }
    case 4:{
        fraction f1, f2;
        printf("Enter numerator and denominator of the first fraction: ");
        scanf("%d %d", &f1.numerator, &f1.denominator);
        printf("Enter numerator and denominator of the second fraction: ");
        scanf("%d %d", &f2.numerator, &f2.denominator);
        int result = compareFractions(f1, f2);
        if (result == 0) {
            printf("The two fractions are equal.\n");
        } else if (result == -1) {
            printf("The first fraction is less than the second fraction.\n");
        } else {
            printf("The first fraction is greater than the second fraction.\n");
        }
```

```
            break;
        }
        case 5:{
            date d;
            printf("Enter the date (dd mm yyyy): ");
            scanf("%d %d %d", &d.day, &d.month, &d.year);
            if (validateDate(&d)) {
                printf("The date is: %d/%d/%d valid\n", d.day, d.month, d.year);
            } else {
                printf("Invalid date entered!\n");
            }
            break;
        }
        case 6:{
            time t1, t2;
            printf("Enter t1 (hr min sec): ");
            scanf("%d %d %d", &t1.hr, &t1.min, &t1.sec);
            printf("Enter t2 (hr min sec): ");
            scanf("%d %d %d", &t2.hr, &t2.min, &t2.sec);
            time sum = addTime(t1, t2);
            time difference = subtractTime(t1, t2);
            printf("Sum: %d:%d:%d\n", sum.hr, sum.min, sum.sec);
            printf("Difference: %d:%d:%d\n", difference.hr, difference.min, difference.sec);
            break;
        }
        default:
            break;
    }
    return 0;

}
```

2. header.h

```
typedef struct student{
   int rollno;
   struct name{
      char firstName[20];
      char middleName[20];
      char lastName[20];
   }name;
   char gender[15];
   struct dob{
      int day;
      int month;
      int year;
   }dob;
   struct marks{
      int maths;
      int science;
      int computerScience;
   }marks;
}student;

typedef struct hotel{
   char name[50];
   char address[100];
   int grade;
   int number_of_rooms;
   int room_charges;
}hotel;

typedef struct time{
   int hr;
```

```
    int min;
    int sec;
}time;

typedef struct fraction{
    int numerator;
    int denominator;
}fraction;

typedef struct date{
    int year;
    int month;
    int day;
}date;

void displayStudent(const student *s);
float calculateAverage(const student *s);
float calculateAggregatePercentage(const student *s);
void hotelsInParticularGrade(const hotel *h, int grade);
void hotelsOfParticularValue(const hotel *h, int value);
void increment_time(time *t);
int compareFractions(fraction f1, fraction f2);
int isLeapYear(int year);
int validateDate(date *d);
time addTime(time t1, time t2);
time subtractTime(time t1, time t2);
```

3. logic.c

```c
#include <stdio.h>
#include "header.h"
#include <string.h>
#include <stdlib.h>

void displayStudent(const student *s){
   printf("Roll Number: %d\n", s->rollno);
   printf("Name: %s %s %s\n", s->name.firstName, s->name.middleName, s->name.lastName);
   printf("Gender: %s\n", s->gender);
   printf("Date of Birth: %02d/%02d/%04d\n", s->dob.day, s->dob.month, s->dob.year);
   printf("Marks:\n");
   printf("  Mathematics: %d\n", s->marks.maths);
   printf("  Science: %d\n", s->marks.science);
   printf("  Computer Science: %d\n", s->marks.computerScience);
}

float calculateAverage(const student *s){
   return (s->marks.maths + s->marks.science + s->marks.computerScience) / 3.0;
}

float calculateAggregatePercentage(const student *s){
   int totalMarks = s->marks.maths + s->marks.science + s->marks.computerScience;
   return (totalMarks / 300.0) * 100;
}

void hotelsInParticularGrade(const hotel *h, int grade){
   if (h->grade == grade){
      printf("Hotel Name: %s\n", h->name);
   }
}
```

```
void hotelsOfParticularValue(const hotel *h, int value){
    if(h->room_charges < value){
        printf("Hotel Name: %s\n", h->name);
    }
}
void increment_time(time *t){
    t->sec++;
    if (t->sec == 60) {
        t->sec = 0;
        t->min++;
        if (t->min == 60) {
            t->min = 0;
            t->hr++;
            if (t->hr == 24) {
                t->hr = 0;
            }
        }
    }
}

int compareFractions(fraction f1, fraction f2){
    float value1 = (float)f1.numerator / f1.denominator;
    float value2 = (float)f2.numerator / f2.denominator;
    if (value1 == value2) {
        return 0;
    } else if (value1 < value2) {
        return -1;
    } else {
        return 1;
    }
}
```

```
int isLeapYear(int year){
   if(year % 400 == 0){
      return 1;
   }
   if(year % 100 == 0){
      return 0;
   }
   if(year % 4 == 0){
      return 1;
   }
   return 0;
}

int validateDate(date *d){
    if (d->month < 1 || d->month > 12) {
      return 0;
   }
   int daysInMonth[] = {31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};
   if (d->month == 2 && isLeapYear(d->year)) {
      daysInMonth[1] = 29;
   }
   if (d->day < 1 || d->day > daysInMonth[d->month - 1]) {
      return 0;
   }
   return 1;
}

time addTime(time t1, time t2){
   time result;
   result.sec = t1.sec + t2.sec;
```

```
    result.min = t1.min + t2.min + (result.sec / 60);
    result.sec %= 60;
    result.hr = t1.hr + t2.hr + (result.min / 60);
    result.min %= 60;
    return result;
}
time subtractTime(time t1, time t2){
    time result;
    result.sec = t1.sec - t2.sec;
    result.min = t1.min - t2.min;
    result.hr = t1.hr - t2.hr;
    if (result.sec < 0) {
        result.sec += 60;
        result.min--;
    }
    if (result.min < 0) {
        result.min += 60;
        result.hr--;
    }
    return result;
}
```